

Question: How would you use NumPy arrays to calculate the average score for each subject and determine the subject with the highest average score? Assume 4x4 matrix that stores marks of each student in given order.

Aim

To use a **NumPy array** to calculate the average marks in each subject and find the subject with the **highest average score**.

Algorithm

1. Start the program.
2. Import the NumPy library.
3. Create a 4×4 **NumPy array** containing marks of 4 students.
4. Arrange the columns in the order:
 - Math
 - Science
 - English
 - History
5. Calculate the average of each column using `np.mean()` with `axis=0`.
6. Store the subject names in a list.
7. Find the position of the highest average using `np.argmax()`.
8. Display the average score of each subject.
9. Display the subject having the highest average.
10. Stop the program.

Python Program :

```
import numpy as np

# 4 students and 4 subjects
# Order: Math, Science, English, History
student_scores = np.array([
    [80, 75, 85, 70],
    [90, 80, 78, 75],
    [70, 85, 90, 80],
    [85, 90, 88, 82]
])

subjects = ["Math", "Science", "English", "History"]

# Calculate average for each subject
average = np.mean(student_scores, axis=0)

# Display average scores
print("Average scores:")

for i in range(4):
    print(subjects[i], "=", average[i])

# Find subject with highest average
```

```
i = np.argmax(average)
```

```
print("\nHighest average subject:", subjects[i])
```

```
print("Highest average:", average[i])
```

Input:

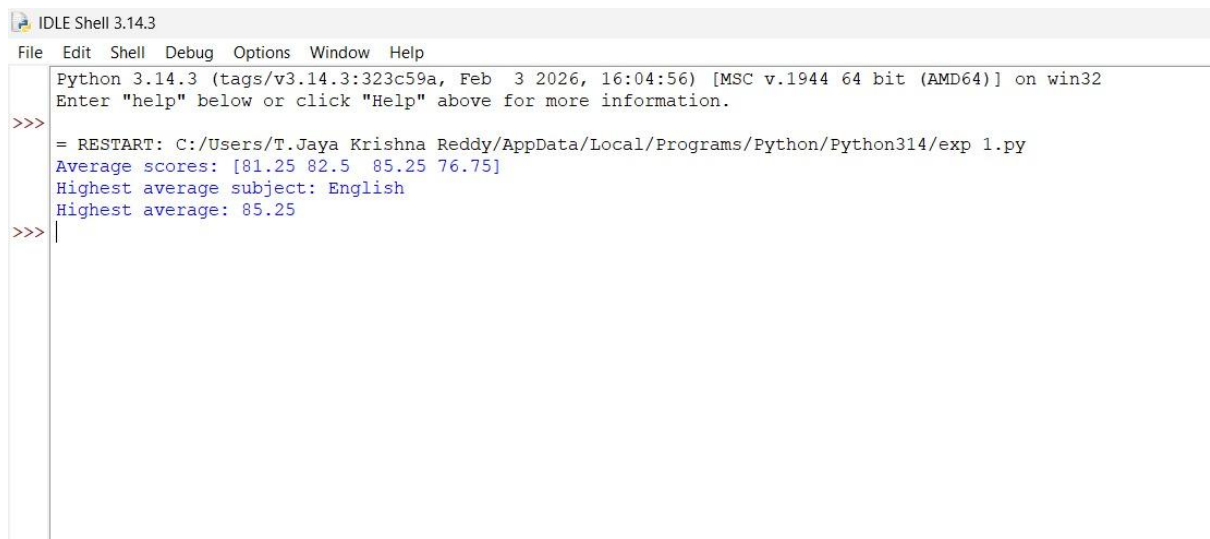
80 75 85 70

90 80 78 75

70 85 90 80

85 90 88 82

Output :



```
IDLE Shell 3.14.3
File Edit Shell Debug Options Window Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/T.Jaya Krishna Reddy/AppData/Local/Programs/Python/Python314/exp 1.py
Average scores: [81.25 82.5 85.25 76.75]
Highest average subject: English
Highest average: 85.25
>>> |
```